L-P-00315D (GSA-FSS)
October 22, 1976
Interim Revision of
Fed. Spec. L-P-315C
May 3, 1972

INTERIM FEDERAL SPECIFICATION

PIPE AND TUBING, NONMETALLIC (PLASTIC)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification L-P-315C, dated May 3, 1972.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plastic pipe and tubing used to carry potable water.

1.2 Classification.

1.2.1 Types, classes, grades, and forms. Pipe and tubing covered by this specification shall be of the following types, classes, grades, and forms as specified (see 6.2):

Type I - Polyethylene (PE) Pipe, Cold Water Only.

Class A - PE 2305.

Grade 1 - Standard Design Ratio (SDR) 9.
Grade 2 - SDR 11.5.

Class B - PE 2306.

Grade 2 - SDR 11.5.
Grade 3 - SDR 15.

Type II - Polybutylene (PB) Hot or Cold Water, SDR 11.

Form p - pipe.
Form t - tubing (see 6.4).

1.2.2 Sizes. Pipe and tubing furnished under this specification shall be of the following nominal pipe and tubing sizes as specified (see 6.2): 1/2, 3/4, 1, 1-1/4, 1-1/2, and 2 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601 - Boxes; Wood, Cleated-Plywood.
PPP-B-621 - Boxes; Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-60 - Tape; Packaging, Waterproof.
PPP-T-97 - Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

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Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 2239 - Specification for Polyethylene (PE) Plastic Pipe (SDR-PR).

D 3309 - Polybutylene (PB) Plastic Hot-Water Distribution System.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Sanitation Foundation Testing Laboratory Inc., (NSF) Standard:

No. 14 - Thermoplastic Materials, Pipe, Fittings, Valves, Joining Materials.

(Application for copies should be addressed to the National Sanitation Foundation Testing Laboratory, Inc., P.O. Box 1468, Ann Arbor, MI 48106.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 General. All pipe and tubing furnished under this specification shall be for transporting potable water.

3.2 Type I. PE pipe shall conform to the requirements of ASTM D 2239 and as specified herein.

3.2.1 Pressure rating. Pressure rating (PR) for PE pipe shall conform to table I.

TABLE I. Pressure ratings for type I pipe

Grade	Class A		Class B	
	p.s.i.	*kPa	p.s.i.	*kPa
1	100	690	---	---
2	80	552	100	690
3	---	---	80	552

*Kilo Pascals

3.3 Type II. PB pipe and tubing shall conform to the requirements of ASTM D 3309 and as specified herein.

3.4 Pipe and tubing for potable water, public health and safety hazards.

3.4.1 Each contractor shall submit to the contracting agency proof that the pipe or tubing, as applicable, which he proposes to supply under this specification conforms to the requirements of the National Sanitation Foundation Standard No. 14. The label, or listing with re-examination, of the National Sanitation Foundation may be accepted as evidence that the pipe or tubing conforms to this requirement.

3.4.2 In lieu of the label, or listing with re-examination, the contractor may submit independent proof satisfactory to the contracting agency, that the pipe or tubing conforms to the applicable requirements of the published standards, including methods of test, of the National Sanitation Foundation Standard No. 14.

3.4.3 Compliance with the above preliminary requirements in regard to public health and safety hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.5 Length. Pipe and tubing shall be furnished in 100, 300, or 500 foot coils as specified (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection. Inspection levels and acceptable quality levels (AQL's) shall be in accordance with MIL-STD-105.

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4.2.1 Dimensional and visual examination of end item. Pipe and tubing shall be examined for the characteristics listed in table II. Examination shall be in accordance with level S-3, using an AQL of 2.5 percent defective for major defects and 6.5 defective for total defects. Dimensions of PE pipe shall be measured in accordance with ASTM D 2239, and those of PB pipe and tubing shall be measured in accordance with ASTM D 3309.

TABLE II. Classification of defects, end item

Defects	Major	Minor
Type, class, grade, form, or size not as specified	X	
Cracks or holes	X	
Foreign inclusions	X	
Dimension not within tolerance	X	
Identification marking omitted, incorrect, illegible		X
Damage or defects affecting function or serviceability	X	
Damage or defects not affecting function or serviceability		X

4.2.2 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The inspection level **shall be S-2 with an AQL of 4.0 percent defective.**

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible, improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

4.3 Test methods.

4.3.1 Type I. PE pipe shall be tested in accordance with ASTM D 2239 (see 3.2).

4.3.2 Type II. PB pipe and tubing shall be tested in accordance with ASTM D 3309 (see 3.3).

4.3.3 Sustained pressure and environmental cracking tests. Due to the length of time required for conducting these tests, they will be required only when the NSF label or listing with re-examination, has been withheld from the pipe or tubing submitted for acceptance by the Government. In the absence of the NSF label or listing with re-examination, the tests for type I pipe shall be conducted in accordance with ASTM D 2239, and tests for type II pipe or tubing shall be in accordance with ASTM D 3309.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A. Plastic pipe with a nominal size of 2 inches or less, shall be coiled to a minimum safe diameter and the coils secured with not less than 5 evenly spaced ties of soft wire, cotton tape, or with 4 ties of tape conforming to PPP-T-97, type III. Prior to coiling, the ends of the coils shall be sealed with plastic plugs or with tape conforming to PPP-T-60, type III, class 1. Noncoiled pipe shall have ends sealed by means that will withstand the hazards of transportation, without damage and will prohibit the entrance of foreign matter to the interior of the pipe.

5.1.2 Level C. The pipe shall be packaged in a manner to provide necessary protection against damage from the supply source to the first receiving activity, except that the ends shall be sealed as specified in 5.1.1.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Pipe of like description shall be packed in boxes conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. The gross weight of the PPP-B-636 box shall not exceed the weight limitations of the box specification and wood boxes shall not exceed 200 pounds. When a single coil of tubing exceeds 250 pounds, each coil packed in a wood box shall be modified by the installation of skids in accordance wtih PPP-B-621.

5.2.2 Level B. Pipe of like description shall be packed as specified in 5.2.1, except the containers shall be the domestic class or type and unless otherwise specified (see 6.2) strapping shall not be required.

5.2.3 Level C. The plastic pipe shall be packed to insure carrier acceptance and safe delivery to destination at lowest rates in compliance with *Uniform Freight Classification rules or National Motor Freight Classification* rules.

5.3 Marking.

5.3.1 Civil agencies. Shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I. PE pipe is primarily intended for conveying potable water at 23°C (73°F) or less with maximum internal pressure shown in table I. It is also used for sprinkler systems, pump installations, air-conditioning water lines, irrigation, and drainage systems.

6.1.2 Type II. PB pipe and tubing is primarily intended for conveying potable water at temperatures up to 82°C (180°F) with a maximum internal pressure of 690 kPa (100 p.s.i.).

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, grade, and form as applicable required (see 1.2.1).
(c) Size required (see 1.2.2).
(d) Length of pipe or tube required (see 3.5).
(e) Level of packaging and packing required (see 5.1 and 5.2).
(f) If strapping is required (see 5.2.2).

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6.3 For civil agency procurements. When level B packing is required by civil agency procurements, the requirements in 5.1.1, level A packaging shall apply.

6.4 The name plastic tubing denotes a particular sizing of plastic pipe in which the outside diameter is essentially the same as the corresponding size of copper tubing. Plastic tubing will have a smaller outside diameter, a smaller inside diameter, and a lesser wall thickness than plastic pipe of the same nominal size and SDR. Plastic tubing will normally be specified for applications in which an outside diameter comparable to that of copper tubing is the determining factor in product selection.

6.5 The relation between standard dimension ratio, hydrostatic design stress, and internal pressure rating is detailed and formulated in ASTM D 3309.

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